

Frequency Support from Electric Vehicles for Advancing Renewable Energy Integration

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Abstract—The integration of renewable energy resources (RERs) in the modern power grid is increasing rapidly because of aggressive decarbonization goals, lower costs, and increased government investment. However, higher penetrations of inverter-based generation can lead to frequency stability issues because of reduced system inertia. This paper develops a framework for quantifying the contribution of electric vehicles (EVs) toward providing frequency support to the grid and thus increasing the penetration limit of renewable energy resources (RERs). EVs are considered to provide both inertial response and primary frequency response support to the grid. A stochastic approach incorporating the uncertainties associated with the behavior of EVs is developed to derive the discharge limit of EV aggregators. A multi-machine system frequency response (MM-SFR) model is developed, which incorporates the dynamic virtual inertia and droop coefficients of EV aggregators derived from the EV control modules. Frequency security constraints are developed from this MM-SFR model, which, along with the converter voltage security and low voltage ride-through constraints, are integrated within a nonlinear optimization framework to determine the RER integration limit. The efficacy of the proposed approach is validated using the RTS-GMLC test system.

Index Terms—Electric vehicles, frequency security, inertia response, renewable energy resources, voltage security.

NOMENCLATURE

A. Sets and indices

$g \in \mathcal{G}$	Set of conventional generators.
$n \in \mathcal{N}$	Set of system nodes or buses.
$(k, l) \in \mathcal{L}$	Set of transmission lines.
$p \in \mathcal{P}$	Set of PV farms, RTPV, and CSP.
$w \in \mathcal{W}$	Set of wind farms.
$s \in \mathcal{S}$	Set of energy storage systems (ESS).
$e \in \mathcal{E}_a$	Set of EV aggregators.

B. Decision variables of NLP

$P_{n,t}^C$	Load curtailment at node n .
$P_{g,t}, P_{p,t}, P_{w,t}$	Power generation of generator g , PV farm p , and wind farm w .
$P_{s,t}^+, P_{s,t}^-$	Charging and discharging power of energy storage system (ESS) s .
$Q_{g,t}$	Reactive power generation of generator g .
$V_{n,t}, \delta_{n,t}$	Voltage magnitude and angle of node n .
β_t	RER integration factor.
$H_{e,t}^{\text{EV}}$	Virtual inertia constant of aggregator e .
$R_{e,t}^{\text{EV},*}$	Droop reciprocal corresponding to QSS frequency for EV aggregator e .
$R_{e,t}^{\text{EV},**}$	Droop reciprocal corresponding to frequency nadir for EV aggregator e .
$P_{e,t}^{\text{set}}, Q_{e,t}^{\text{set}}$	Active and reactive power setpoints for EV aggregator e .

C. MMSFR parameters

H_g	Inertia of generator g .
D_g	Damping coefficient of generator g .
K_g	Droop gain of generator g .
R_g	Droop coefficient of generator g .
T_g	Governor time constant of generator g .
F_g	Fraction of power from high-pressure turbines of generator g .
ΔP_t	Disturbance size at time t .
$\gamma_{g,t}$	Dispatch status of generator g obtained using the capacity flow model.
$P_g^{\text{max}}, P_g^{\text{min}}$	Maximum and minimum generation capacity of conventional generator g .
ξ_t	System damping ratio for time t .
ω_t	Natural frequency for time t .
τ_t^{max}	Time of frequency nadir.

D. EV aggregator and individual EV parameters

f_r	Rated frequency (60 Hz in this paper).
K_e^{IR}	IR gain of the e^{th} aggregator.
R_e^{EV}	Rated droop of the e^{th} aggregator.
N_{EV}	Total number of EVs in the system.
N_t^{ch}	Total number of EVs in the charging station at time t .
$\text{SoC}_{i,t-1}^{\text{ev}}$	State-of-charge of i^{th} EV at previous time-step.
$P_{i,t}^{\text{ev},+}, P_{i,t}^{\text{ev},-}$	Charging and discharging power of i^{th} EV.
$\frac{P_{i,t}^{\text{ev},+}, P_{i,t}^{\text{ev},-}}{B_i^{\text{ev}}, B_i^{\text{ev}}}$	Charging and discharging limits of i^{th} EV.
$B_i^{\text{ev}}, B_i^{\text{ev}}$	Maximum and minimum battery capacity of i^{th} EV.
$B_{i,t}^{\text{ev}}$	Remaining battery capacity of the i^{th} EV.
$\eta_i^{\text{ev},+}, \eta_i^{\text{ev},-}$	EV charging and discharging efficiencies.
$d_{i,t}^{\text{ev}}$	Distance traveled (in miles) by the i^{th} EV.
c_i^{ev}	Energy consumed per mile.
$L_{e,t}^{\text{EV}}$	Charging load imposed by aggregator e .
$P_{e,t}^{\text{EV}}$	Aggregated discharge limit of aggregator e .
$B_{e,t}^{\text{EV}}$	Total energy capacity of aggregator e .
$E_{e,t}, \delta_{e,t}$	Voltage and angle at converter PCC.
x_g	Reactance of line connecting PCC and grid.

E. Other parameters in NLP

λ_c	Large multiplier to penalize curtailed load.
c_s^+, c_s^-	Cost coefficient of s^{th} ESS charging and discharging.
$P_{n,t}^D, Q_{n,t}^D$	Node n active and reactive power demand.
$G_{n,i}, B_{n,i}$	Real and imaginary component of bus admittance matrix component $Y_{n,i}$.
$V_n^{\text{min}}, V_n^{\text{max}}$	Minimum and maximum voltage limits at node n .

$u_{g,t}, u_{kl,t}$	Forced outage status (ON/OFF) of generator g and transmission line kl .
$P_{kl}^{\min}, P_{kl}^{\max}$	Power limits of transmission line connecting buses k and l .
$ Y_{k,l} , \theta_{k,l}$	Magnitude and angle of admittance matrix component $Y_{k,l}$.
$P_{p,t}^{\max}, P_{w,t}^{\max}$	Maximum generation limits of PV farm p and wind farm w .
$\overline{P_s^+}, \overline{P_s^-}$	Charging and discharging limit of ESS s .
$\overline{E_s^+}, \overline{E_s^-}$	Minimum and maximum energy of ESS s .
η_s^+, η_s^-	Charging and discharging efficiencies of ESS s .
ΔV_e	Voltage sag at e^{th} EV aggregator bus.
$\left \frac{df}{d\tau} \right _{\lim}$	RoCoF safety limit.
$\Delta f^{ss, \lim}$	QSS frequency safety limit.
$\Delta f^{\lim}$	Frequency nadir safety limit.
$\frac{B_{e,t}^{EV}}{\rho}$	Lower capacity limit of EV aggregator e .
ρ	Approximate ratio of frequency nadir and QSS frequency.
$\eta^{EV, -}$	Discharge efficiency of aggregator e .

I. INTRODUCTION

A. Background

Propelled by low-carbon energy policies, low costs of operation, and the goal of phasing out fossil fuels, the modern grid is experiencing rapid integration of renewable energy resources (RERs). The increasing penetration of RERs by replacement of conventional generation can lead to increased frequency stability issues in the grid due to their intermittent and low-inertia characteristics [1]. This issue may lead to system operators curtailing RER generation and planners investing in less RER-based generation, thus delaying the transition to a decarbonized grid. A potential solution to this problem can be utilizing EVs to provide frequency support to the grid. The integration of EVs has been increasing globally and is projected to increase from 30 million in 2022 to about 240 million in 2030 [2]. EVs are capable of charging and discharging at high power levels to provide inertia support to the grid. This capability is being enhanced as the power electronic converters of EVs improve. This makes EVs good candidates for providing frequency support to the grid, leading to increased penetration of RERs and thus accelerating the transition to a decarbonized future.

B. Relevant Literature & Research Gaps

The contribution of EVs toward frequency stability using virtual inertia has been investigated extensively in recent years. Among the different techniques used to extract frequency support from EVs, a popular approach is the inclusion of EVs in the system load frequency control (LFC) model. Plug-in EVs are included in the LFC model using derivative control-based inertia emulators with tunable gain parameters in the approaches developed in [3], [4]. Another widely used approach for virtual inertia emulation is using different control strategies to generate converter switching signals for the primary converter in grid-to-vehicle connections [5], [6].

Recently, model predictive control has been used in virtual inertia emulation due to its robustness under RER uncertainties [7], [8]. In addition to the inertia response (IR) based on system RoCoF, EVs can also support the grid in the frequency recovery phase. Primary frequency response (PFR) using EVs can be achieved using a droop gain parameter controlling the amount of EV power injection after the V2G aggregators determine the discharging/charging limits [9]–[12]. Research in [9], [10] propose models for EVs to provide primary frequency support under contingencies whereas [11], [12] illustrate the potential frequency support from EVs in real-world grids. The aforementioned works that model the IR and PFR support from EVs deploy simulation models to assess the frequency response. While the simulation-based models can capture diverse operational characteristics, they are unsuitable for planning analysis, where simulations are performed over large time horizons and require solving optimization problems.

Along with accurate modeling of system frequency dynamics under EV integration, a significant interest of the research community lies in leveraging the frequency support capabilities of EVs and converter-based generation to enhance RER penetration. The simulation-based models mentioned above mostly focus on assessing the frequency response with IR and PFR support from EVs but do not assess the contribution of EVs on the integration limit of RERs. References [13]–[16] propose approaches to directly evaluate the integration limits of RERs and RER-based microgrids in bulk power systems with respect to the system frequency response. These works, however, do not incorporate the frequency support from EVs or other converter-based generators into their models.

Frequency security-constrained stochastic unit commitment (FSCUC) frameworks proposed in [17]–[19] incorporate the frequency support from converter-based resources using the multi-machine system frequency response model. They model the IR from converter-based resources using the virtual synchronous machine control, while the PFR is modeled using the droop control. Also, these works implicitly enforce a limit on RER integration using the frequency security constraints. However, these studies deploy constant values for virtual inertia and droop gains, which leads to an overestimated support from converter-based generation under large disturbances [20]. Recent works in [20], [21] implement virtual inertia parameters dependent on the unit-commitment decision variables while [22], [23] deploy variable droop gain parameters in their FSCUC models to incorporate the frequency support from converter-based generation more accurately. These works either only consider the IR [20], [21] or only the PFR [20], [21] from converter-based resources in their FSCUC models. In addition, while the works mentioned consider various aspects of frequency support from converter-based resources, they do not consider the voltage stability or security constraints on the converter interface. This is a major drawback of existing literature since the step disturbances considered for frequency stability analysis are typically accompanied by voltage sags at the converter point of connection. To ensure converter stability under reduced grid voltage, the converters must readjust the active and reactive power setpoints [24]. The readjusted setpoints affect the frequency support from converter-based

resources and should be considered while estimating the RER penetration limits. The research in [25]–[27] addresses the issue of steady-state voltage stability by incorporating the static voltage stability constraints and optimizing the active and reactive power outputs of converter-based resources as decision variables in the FSCUC framework. However, the stability constraints in [25]–[27] are limited to steady-state and do not ensure converter stability and low voltage ride-through (LVRT) compliance under grid voltage sags. In addition, none of the works mentioned incorporate EV aggregators as a potential converter-based resource in their FSCUC formulations.

C. Summary of Contributions

This paper develops a framework that not only quantifies the frequency support from EVs but also derives an optimization framework for calculating RER penetration limits and demonstrates how frequency support from EVs can enable increased RER penetration. The contribution of EVs to system inertia and regulation is modeled using a modified multi-machine system frequency response (MM-SFR) model. The virtual inertia and droop gain of the EV aggregators are modeled as decision variables, and their expressions in terms of the RER integration limit are derived using the control modules for IR and PFR support. The penetration limit of RERs is then obtained using a non-linear programming (NLP) optimization framework that incorporates the AC power flow constraints in addition to the IR and PFR support constraints of EV aggregators, frequency security constraints, converter voltage security, and LVRT constraints formulated using the requirements set forth by the IEEE-Standard 2800-2022 [28]. Dynamic simulations, 24-hour dispatches, and large time-horizon stochastic simulations are performed to demonstrate the efficacy of the proposed approach. The major contributions of the paper are listed as follows:

- 1) Developing an improved MM-SFR model that incorporates frequency support from plug-in EVs using dynamic virtual inertia and droop control of EV aggregators. We derive expressions for the EV inertia and droop parameters in terms of the RER penetration limit, IR, and PFR control parameters and deploy them into the MM-SFR model to obtain the revised frequency security constraints.
- 2) Formulating a NLP framework to determine the maximum penetration limit of RERs while incorporating the frequency support capabilities of EV aggregators. The framework is comprehensive as it considers the ACOF constraints, IR and PFR constraints of EV aggregators, converter voltage security constraints, and LVRT constraints while evaluating the frequency support from EV

aggregators. Existing literature considers a subset of these constraints while evaluating the RER penetration limit.

- 3) Developing a stochastic approach to quantify the discharge capacity of EV aggregators for providing frequency support, a step not within the scope of existing works that consider frequency support from converter-based generation. The proposed approach uses probabilistic techniques to aggregate the charge and discharge characteristics of individual EVs and derives the discharge limits of EV aggregators.

D. Organization

The remainder of the paper is organized as follows. Section II presents the MM-SFR model incorporating frequency support from EVs along with the models for IR and PFR support from EVs. Section III presents the random sampling approach to determine the total number of connected EVs. Section IV develops the converter voltage security criterion. Section V develops a NLP framework to determine the RER penetration limit. It is followed by simulation and results in section VI and conclusion in section VII.

II. FORMULATION OF FREQUENCY SECURITY CONSTRAINTS AND IR, PFR CONSTRAINTS OF EV AGGREGATORS

This section describes the MM-SFR model deployed in this work to incorporate the participation of EV aggregators for frequency support. This model has been derived by improving the basic model proposed in [29]. The modified MM-SFR model is illustrated in Fig. 1.

From Fig. 1, the system frequency deviation under a step load disturbance ΔP_t is expressed as equation (1). Assuming that the governor time constants for all conventional generators are identical $T_g = T_r$, and the EV time constants are 2-3 orders of magnitude smaller than that of conventional generators $T_r \gg T_e^{EV} = 0$ [17], the droop component of the EVs is added to the conventional generator equivalent damping D_t^{CV} . The resulting expression in (1) can then be simplified as follows:

$$\Delta f_t(s) = \frac{\Delta P_t}{s \left(D_t^T + 2H_t^T s + \sum_{g \in \mathcal{G}} \frac{K_g(1+F_g T_r s)}{R_g(1+T_r s)} \right)} \quad (2)$$

The equivalent damping and inertia constants in (2) are expressed as follows:

$$D_t^T = \beta_t D_t^{CV} + D_t^{EV} = \beta_t \sum_{g \in \mathcal{G}} \gamma_{g,t} D_g P_g^{\max} + D_t^{EV}$$

$$H_t^T = \beta_t H_t^{CV} + H_t^{EV} = \beta_t \sum_{g \in \mathcal{G}} \gamma_{g,t} H_g P_g^{\max} + H_t^{EV}$$

$$\Delta f_t(s) = \frac{\Delta P_t}{s \left(\underbrace{D_t^{CV} + 2H_t^{CV} s + \sum_{g \in \mathcal{G}} \left(\frac{K_g(1+F_g T_r s)}{R_g(1+T_g s)} \right)}_{\text{conventional generators}} + \underbrace{2s \sum_{e \in \mathcal{E}_a} H_{e,t}^{EV}}_{\text{EV IR control}} + \underbrace{\sum_{e \in \mathcal{E}_a} \left(\frac{1}{R_{e,t}^{EV}(1+T_e^{EV} s)} \right)}_{\text{EV PFR control}} \right)} \quad (1)$$

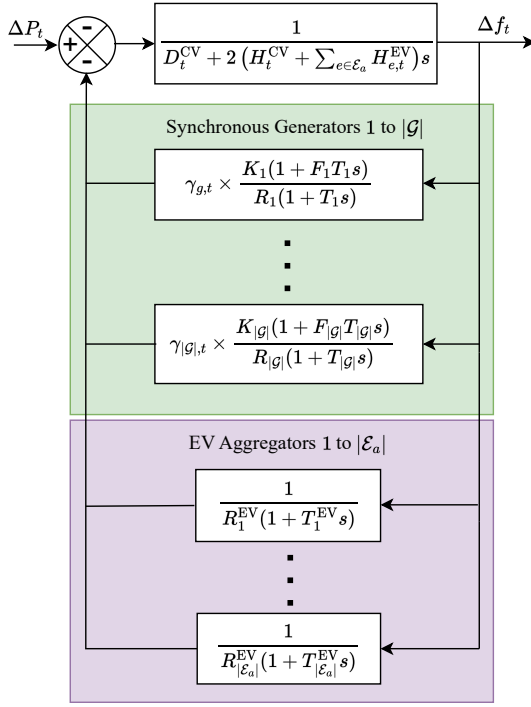


Fig. 1: LFC model for a multi-machine system with EVs.

In the equations above, the RER integration factor β_t denotes the level of conventional generator displacement by RERs. It is a continuous decision variable that takes values between 0 and 1, with $\beta_t = 0$ meaning all the conventional generators are displaced by RERs. The frequency deviation in (2) can then be expressed as follows:

$$\Delta f_t(s) = \frac{\Delta P_t}{2H_t^T T_r} \frac{1 + T_r s}{s(s^2 + 2\xi_t \omega_t s + \omega_t^2)} \quad (3)$$

Applying Laplace transformation to (3), the time-domain frequency deviation can be derived as:

$$\Delta f_t(\tau) = \frac{\Delta P_t}{2H_t^T T_r \omega_t^2} \left(1 - \frac{1}{\sqrt{1 - \xi_t^2}} e^{-\xi_t \omega_t \tau} \cos(\omega_t \sqrt{1 - \xi_t^2} \tau - \phi_t) \right) + \frac{\Delta P_t}{2H_t^T \omega_t \sqrt{1 - \xi_t^2}} e^{-\xi_t \omega_t \tau} \cos(\omega_t \sqrt{1 - \xi_t^2} \tau) \quad (4)$$

where:

$$\xi_t = 0.5 \times \frac{2H_t^T + T_r(D_t^T + \beta_t F_t^T)}{\sqrt{2H_t^T T_r(D_t^T + \beta_t R_t^T)}}, \quad \omega_t = \sqrt{\frac{D_t^T + \beta_t R_t^T}{2H_t^T T_r}},$$

$$F_t^T = \sum_{g \in \mathcal{G}} \frac{\gamma_{g,t} K_g F_g P_g^{\max}}{R_g}, \quad R_t^T = \sum_{g \in \mathcal{G}} \frac{\gamma_{g,t} K_g P_g^{\max}}{R_g}.$$

A. System RoCoF with RERs and IR Support of EVs

The expression for system rate of change of frequency (RoCoF) is obtained by differentiating equation (4) with respect

to τ . The system attains maximum RoCoF at $\tau = 0^+$ and it is expressed as follows:

$$\left| \frac{df}{d\tau} \right|_t^{\max} = \frac{\Delta P_t}{2H_t^T} = \frac{\Delta P_t}{2(\beta_t H_t^{\text{CV}} + H_t^{\text{EV}})} \quad (5)$$

It can be noticed that the RoCoF constraint is dependent on the RER penetration limit β_t and the EV virtual inertia H_t^{EV} . With a larger penetration of RERs (smaller β_t), the system inertia reduces, and RoCoF increases. However, the EV aggregators alleviate this issue with the virtual inertia support in the form of H_t^{EV} . The total inertia from all EV aggregators is expressed as follows:

$$H_t^{\text{EV}} = \sum_{e \in \mathcal{E}_a} H_{e,t}^{\text{EV}} P_{e,t}^{\text{EV}} \quad (6)$$

The total inertia from EV aggregators is dependent on the virtual inertia constant $H_{e,t}^{\text{EV}}$ and the discharge capability $P_{e,t}^{\text{EV}}$ of all EV aggregators. In this work, instead of deploying static values of $H_{e,t}^{\text{EV}}$, they are set as decision variables in the optimization framework, and their values are updated dynamically along with the RER integration limit. The derivative control strategy in Fig. 2 is used to model the IR support from EVs and establish the relation between the EV virtual inertia constant and the RER integration factor.

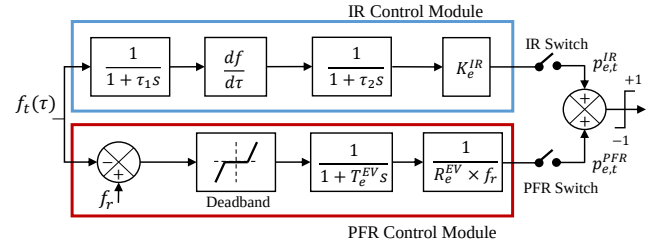


Fig. 2: Aggregated EVs participating in IR and PFR.

During the maximum RoCoF at $\tau = 0^+$, the PFR switch in Fig. 2 can be assumed to be open as the frequency deviation is near zero. As a result, the power output of an EV aggregator e during this time period can be solely attributed to the IR power signal $p_{e,t}^{\text{IR}}$. Reformulating the swing equation, the relation between the virtual inertia constant and IR power for the EV aggregator e at time t can be expressed as follows [30]:

$$H_{e,t}^{\text{EV}} = p_{e,t}^{\text{IR}} \frac{f_r}{2} \left(\left| \frac{df}{d\tau} \right|_t^{\max, \text{cv}} \right)^{-1} \quad (7)$$

where $\left| \frac{df}{d\tau} \right|_t^{\max, \text{cv}}$ is the maximum RoCoF for dispatch hour t without frequency support from EVs. Within the analytical expression of the virtual inertia constant representing the IR loop in equation (7), reference [30] uses a simulation-based approach to determine the maximum RoCoF term $\left| \frac{df}{d\tau} \right|_t^{\max, \text{cv}}$. However, since the virtual inertia constant needs to be included in the NLP framework in the proposed work, using a simulation model is impractical as it cannot be incorporated into the MM-SFR model. Instead, we derive the virtual inertia constant in terms of the renewable integration factor, which is a decision variable in the proposed work. From Fig. 2, the

reference signal for IR corresponding to maximum RoCoF is obtained as follows:

$$p_{e,t}^{\text{IR}} = K_e^{\text{IR}} \frac{1}{(1 + s\tau_1)(1 + s\tau_2)} \left| \frac{df}{d\tau} \right|_t^{\text{max,cv}} \quad (8)$$

Two cases exist as follows:

Case I: When $p_{e,t}^{\text{IR}} < 1$, plugging the expression of $p_{e,t}^{\text{IR}}$ into (7) and ignoring the filtering time constants, the EV inertia constant is equal to the rated value and it is expressed as:

$$H_{e,t}^{\text{EV}} = H_{e,t}^{\text{EV, rated}} = K_e^{\text{IR}} \frac{f_r}{2} \quad (9)$$

Case II: When $p_{e,t}^{\text{IR}} \geq 1$, which can occur due to high RoCoF or a high K_e^{IR} setting, the saturation block limits $p_{e,t}^{\text{IR}}$ to 1. In this case, $H_{e,t}^{\text{EV}}$ is expressed as follows:

$$H_{e,t}^{\text{EV}} = \frac{f_r}{2} \left(\left| \frac{df}{d\tau} \right|_t^{\text{max,cv}} \right)^{-1} \quad (10)$$

Expressing the EV virtual inertia constant in (10) in terms of the RER integration factor:

$$H_{e,t}^{\text{EV}} = \frac{f_r \beta_t H_t^{\text{CV}}}{\Delta P_t} \quad (11)$$

Hence, the inertia constant for EV aggregator e can be approximated as follows:

$$H_{e,t}^{\text{EV}} = \min \left(K_e^{\text{IR}} \frac{f_r}{2}, \frac{f_r \beta_t H_t^{\text{CV}}}{\Delta P_t} \right) \quad (12)$$

From the above equation and Fig. 3, it can be observed that the virtual inertia constants $H_{e,t}^{\text{EV}}$ of EV aggregators are dependent on static parameters such as the IR gain factor, rated frequency, and also the dynamic factors like the RER integration factor, which is a decision variable in this work. If static values of EV inertia constant are taken irrespective of the RER integration, then the IR contribution of EVs is overestimated, especially when β_t is small. Hence, in this work, $H_{e,t}^{\text{EV}}$ are treated as decision variables and are evaluated within the optimization framework in section V.

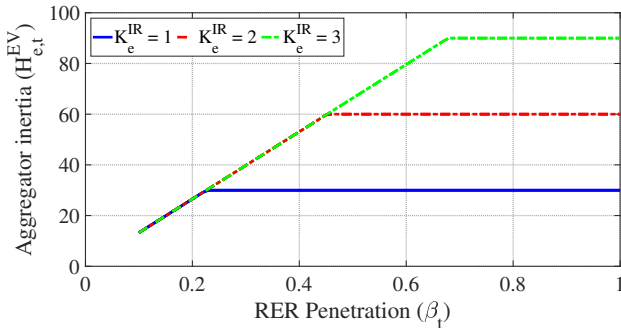


Fig. 3: Effective inertia constant for aggregated EVs with different values of IR gain.

B. QSS Frequency with RERs and PFR Support of EVs

The quasi-steady-state (QSS) frequency deviation can be estimated by using the final value theorem on the frequency deviation expression in equation (3). It is expressed as follows:

$$\Delta f_t^{\text{ss}} = \lim_{s \rightarrow 0} s \Delta f_t(s) = \frac{\Delta P_t}{\beta_t (R_t^T + D_t^{\text{CV}}) + D_t^{\text{EV}}} \quad (13)$$

It can be noticed that the system QSS frequency is dependent on the RER integration limit β_t , and the PFR contribution of all EV aggregators in the form of D_t^{EV} . The expression for D_t^{EV} can be further expanded as follows:

$$D_t^{\text{EV}} = \sum_{e \in \mathcal{E}_a} \frac{P_{e,t}^{\text{EV}}}{R_{e,t}^{\text{EV}}} = \sum_{e \in \mathcal{E}_a} P_{e,t}^{\text{EV}} R_{e,t}^{\text{EV}*} \quad (14)$$

The PFR support from the EV aggregators is modeled using the PFR control loop in Fig. 2. As opposed to the IR control that acts on the system RoCoF, the PFR control loop acts on the frequency deviation. At the QSS frequency, the IR switch in Fig. 2 can be assumed to be open as the RoCoF is zero at steady-state. The value of $R_{e,t}^{\text{EV}}$ is determined using the rated value of droop constant, deadband setting, and the current QSS frequency. This effect is illustrated in Fig. 4 which shows the change in equivalent droop with the system QSS frequency.

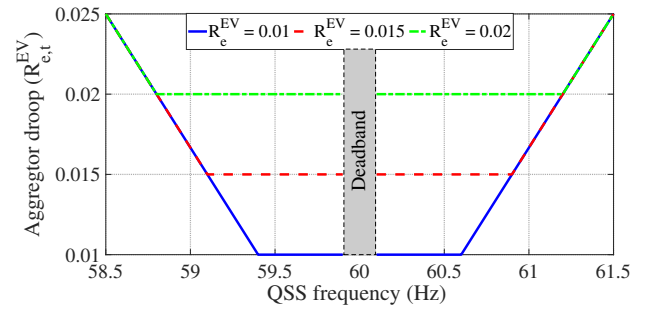


Fig. 4: Effective droop value for aggregated EVs with different values of rated droop and QSS frequency.

Two cases exist as follows:

Case I: After the deadband, a constant droop provides PFR until $p_{e,t}^{\text{PFR}} \leq 1$ as seen in Fig. 4. The value of $p_{e,t}^{\text{PFR}}$ is less than 1 until the actual QSS frequency is smaller than the rated QSS frequency. The EV droop constant for this case is equal to the rated droop, i.e. $R_{e,t}^{\text{EV}} = R_e^{\text{EV}}$.

Case II: If the QSS frequency is larger than the rated QSS frequency, the effective droop increases linearly since $p_{e,t}^{\text{PFR}}$ saturates to 1 p.u. Ignoring the time constant, the EV droop constant for this case can be expressed as follows:

$$R_{e,t}^{\text{EV}} = \Delta f_t^{\text{ss}} \quad (15)$$

The value of the droop constant in case II is dependent on the QSS frequency Δf_t^{ss} , which in turn depends on the RER integration, as seen in equation (13). Replacing the droop constant $R_{e,t}^{\text{EV}}$ by its reciprocal $R_e^{\text{EV}*}$ and plugging equations (13) and (14), equation (15) can be expressed as:

$$R_{e,t}^{\text{EV}*} = \frac{\beta_t (R_t^T + D_t^{\text{CV}}) + \sum_{i \in \mathcal{E}_a, i \neq e} P_{i,t}^{\text{EV}} R_{i,t}^{\text{EV}*}}{\Delta P_t - P_{e,t}^{\text{EV}}} \quad (16)$$

In summary, the droop reciprocal $R_e^{\text{EV}*}$ of the e^{th} EV aggregator during QSS frequency is expressed as follows:

$$R_{e,t}^{\text{EV}*} = \min \left(\frac{1}{R_e^{\text{EV}}}, \frac{\beta_t (R_t^T + D_t^{\text{CV}}) + \sum_{i \in \mathcal{E}_a, i \neq e} P_{i,t}^{\text{EV}} R_{i,t}^{\text{EV}*}}{\Delta P_t - P_{e,t}^{\text{EV}}} \right) \quad (17)$$

From equation (16), it is clear that as the RER integration level increases (β_t decreases) or the disturbance size increases, the droop reciprocal $R_{e,t}^{\text{EV},*}$ of the e^{th} aggregator decreases. This effect is counteracted by the PFR contribution from other EV aggregators (second term in the numerator) and the discharge capability of the e^{th} EV aggregator $P_{e,t}^{\text{EV}}$. Since the droop reciprocals depend on other decision variables, they are treated as decision variables, and their values are updated dynamically with the changing RER integration and EV discharge capabilities.

C. Frequency Nadir in the presence of RERs and EVs

For a given disturbance level ΔP_t , the maximum frequency deviation (nadir) can be estimated at the first zero crossing of the RoCoF. Hence, taking the derivative of (4) and equating it to zero, the frequency nadir can then be expressed as:

$$\Delta f_t^{\text{max}} = \frac{\Delta P_t}{\beta_t R_t^T + D_t^T} \left(1 + e^{-\xi_t \omega_t \tau_t^{\text{max}}} \sqrt{\frac{T_r \beta_t (R_t^T - F_t^T)}{2 H_t^T}} \right) \quad (18)$$

where,

$$D_t^T = \beta_t \sum_{g \in \mathcal{G}} \gamma_{g,t} D_g P_g^{\text{max}} + \sum_{e \in \mathcal{E}_a} P_{e,t}^{\text{EV}} R_{e,t}^{\text{EV},**}$$

$$\tau_t^{\text{max}} = \frac{1}{\omega_t \sqrt{1 - \xi_t^2}} \tan^{-1} \left(\frac{\omega_t \sqrt{1 - \xi_t^2}}{\xi_t \omega_t - 1/T_r} \right)$$

All the other parameters are the same as equation (4). The joint contribution of IR and PFR from EV aggregators helps to arrest the frequency nadir. As a result, during frequency nadir, both the IR and PFR switches are assumed to be in a closed state in Fig. 2. The expression for the EV virtual inertia constant in equation (12) can be used for frequency nadir as it specifies a strict lower bound on the IR contribution. However, the lower bound of the droop reciprocal for QSS frequency in (13) is not applicable for frequency nadir. Since the frequency nadir is always greater than the QSS frequency, using the value of $R_{e,t}^{\text{EV},*}$ in equation (18) overestimates the PFR contribution of EV aggregators. Hence, we propose a droop reciprocal for frequency nadir which is expressed as follows:

$$R_{e,t}^{\text{EV},**} = \min \left(\frac{1}{R_e^{\text{EV}}}, \frac{\beta_t (R_t^T + D_t^{\text{CV}}) + \sum_{i \in \mathcal{E}_a, i \neq e} P_{i,t}^{\text{EV}} R_{i,t}^{\text{EV},**}}{\rho \Delta P_t - P_{e,t}^{\text{EV}}} \right) \quad (19)$$

where ρ is an approximate ratio of frequency nadir to QSS frequency (always greater than 1).

Using equations (5), (13), and (18), the frequency security constraints can be formulated to assess the safe penetration limit of RERs. By setting the frequency security limits to be equal to or less than the safety limits in the optimization framework, the optimal value of RER integration limit to ensure frequency security can be determined. Also, the aforementioned equations show that the frequency nadir, maximum RoCoF, and QSS frequency are directly dependent on the RER integration factor β_t , the EV aggregator virtual inertia constant $H_{e,t}^{\text{EV}}$, and droop reciprocals $R_{e,t}^{\text{EV},*}$, $R_{e,t}^{\text{EV},**}$. Hence,

the expressions in equations (12), (17), and (19) are used to enforce the IR and PFR constraints for EV aggregators. It can also be observed that the IR and PFR contribution from an EV aggregator e is dependent on the discharge capacity $P_{e,t}^{\text{EV}}$. The discharge capacity of the EVs depends on the total number of vehicles in the charging state, i.e., the number of vehicles at the charging stations/EV aggregators at the particular time step. In the next section, a stochastic procedure to determine the discharge capacity of EV aggregators is proposed.

III. DISCHARGE CAPACITY OF EV AGGREGATORS

In this work, we consider the plug-in EVs to have Vehicle-to-Grid (V2G) technology. V2G enables EVs to respond rapidly to frequency fluctuations by feeding surplus power into the grid or drawing power as needed [8]. With V2G, the EVs are equipped with bidirectional chargers, allowing them to charge from the grid and discharge power back into it. In addition to V2G, while considering frequency support from a massive number of EVs, it is paramount to model the actual number of EVs participating in frequency support. If all the EVs in the system are assumed to participate in frequency regulation at any time step, the frequency support from EVs is significantly overestimated. Hence, in this work, only the EVs that are already at the charging stations are assumed to be able to provide frequency support. To model the number of EVs present at the charging stations and eligible to provide frequency support, we have developed a stochastic procedure in Algorithm 1. For each EV, the cumulative density function of the EV SoC at the charging start time extracted from [31] is compared against a random number from the uniform distribution. Then, depending on the operation modes of EVs, the actual number of EVs at the charging station, their charging load, and their discharge capability for frequency support are estimated. For EVs not at the charging station, their traveling distance is randomly assigned using the NHTS database [32].

Algorithm 1 Operation of plug-in EVs at time-step t .

```

1:  $i \leftarrow 1$ 
2:  $N_t^{\text{ch}} \leftarrow 0$ 
3: while  $i \leq N_{\text{EV}}$  do
4:   if  $\text{rand}[\mathcal{U}(0,1)] > \text{cdf}(\text{SoC}_{i,t-1}^{\text{ev}})$  then
5:     Set operation mode of EV  $i$  to charging.
6:      $P_{i,t}^{\text{ev},+} \leftarrow \min(P^{\text{ev},+}, (B_i^{\text{ev}} - B_{i,t-1}^{\text{ev}})/\Delta t)$ 
7:      $B_{i,t}^{\text{ev}} \leftarrow B_{i,t-1}^{\text{ev}} + \eta^{\text{ev},+} P_{i,t}^{\text{ev},+} \Delta t$ 
8:      $N_t^{\text{ch}} \leftarrow N_t^{\text{ch}} + 1$ 
9:     if frequency support required then
10:      Set operation mode of  $i$  to grid support.
11:       $P_{i,t}^{\text{ev},-} \leftarrow \min(P^{\text{ev},-}, (B_{i,t-1}^{\text{ev}} - B_i^{\text{ev}})/\Delta t)$ 
12:    end if
13:  else
14:    Set operation mode of EV  $i$  to discharging.
15:    Determine  $d_{i,t}^{\text{ev}}$  using the NHTS travel database.
16:     $B_{i,t}^{\text{ev}} \leftarrow B_{i,t-1}^{\text{ev}} - d_{i,t}^{\text{ev}} \times c^{\text{ev}}/\eta^{\text{ev},-}$ 
17:  end if
18:   $i \leftarrow i + 1$ 
19: end while

```

In the charging state, the EVs connected to the grid consume power to charge their batteries. The charging powers for all the EVs in the charging state are summed to obtain the load imposed by the aggregator e as follows:

$$L_{e,t}^{\text{EV}} = \sum_{i=1}^{N_t^{\text{ch}}} P_{i,t}^{\text{ev},+} \quad (20)$$

Similarly, in the discharge mode, the EV is not connected to the grid and discharges based on the randomly selected traveling distance for the given hour. In the event of a disturbance, the EVs that are currently connected to the charging stations are in “grid support” mode and provide IR and PFR services. By summing the discharge capacities of all EVs connected to the aggregator e for a particular hour, the discharge power and energy limits for EV aggregator e are obtained as follows:

$$P_{e,t}^{\text{EV}} = \sum_{i=1}^{N_t^{\text{ch}}} P_{i,t}^{\text{ev},-}, \quad B_{e,t}^{\text{EV}} = P_{e,t}^{\text{EV}} \Delta t \quad (21)$$

While this value of discharge limit can be used to evaluate the frequency support, the converter voltage security criterion dictates that the converter’s active and reactive power setpoints must be altered depending on the grid voltage, further constraining the value of $P_{e,t}^{\text{EV}}$. Hence, the converter voltage security constraint is developed in the following section that constrains the EV aggregator discharge capacity to obtain the actual frequency support capability of EV aggregators.

IV. CONVERTER VOLTAGE SECURITY CONSTRAINTS

In this work, EV aggregators are assumed to interact with the grid as grid-feeding converters. A brief overview of the operation of grid-feeding converters is presented in this paper. Readers are referred to [24] for detailed operation of grid-feeding converters. Fig. 5 illustrates the structure of grid-feeding converters used in this work.

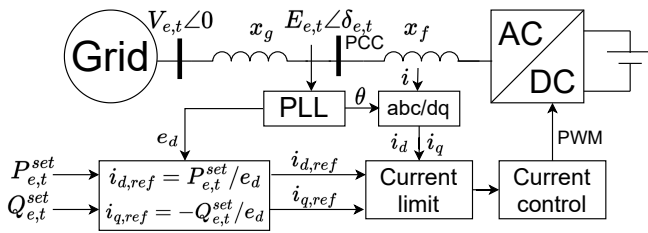


Fig. 5: Control strategy for grid-feeding converter e .

The synchronization with the point of common coupling (PCC) is achieved via a phase-locked loop (PLL) where the detected phase θ tracks the PCC phase $\delta_{e,t}$. Since we assume continuous synchronization, the q-axis voltage is 0 while $e_d = \sqrt{2}E_{e,t}$. Due to this, the active and reactive power are decoupled, and the power flow to the grid is equal to the active and reactive power setpoints. Since we assume the grid voltage as a reference, the phase of $V_{e,t}$ is assumed to be zero. The active and reactive converter-to-grid power are as follows:

$$P_{e,t}^{\text{conv}} = \frac{E_{e,t}V_{e,t}}{x_g} \sin(\delta_{e,t}), \quad Q_{e,t}^{\text{conv}} = \frac{E_{e,t}^2}{x_g} - \frac{E_{e,t}V_{e,t}}{x_g} \cos(\delta_{e,t}) \quad (22)$$

Eliminating $\delta_{e,t}$ in (22), the PCC voltage with respect to the active and reactive powers (PV and VQ curves) are:

$$E_{e,t} = \sqrt{\frac{V_{e,t}^2}{2} + Q_{e,t}^{\text{conv}} x_g} \pm \sqrt{\frac{V_{e,t}^4}{4} - (P_{e,t}^{\text{conv}})^2 x_g^2 + Q_{e,t}^{\text{conv}} x_g V_{e,t}} \quad (23)$$

From the equation above, the maximum deliverable power from the converter to the grid is obtained as follows:

$$\overline{P}_{e,t}^{\text{conv}} = \frac{1 + \sin(\phi_{e,t})}{\cos(\phi_{e,t})} \frac{V_{e,t}^2}{2x_g} \quad \text{where,} \quad \phi_{e,t} = \tan^{-1} \left(\frac{Q_{e,t}^{\text{conv}}}{\overline{P}_{e,t}^{\text{conv}}} \right) \quad (24)$$

As the grid voltage reduces, the value of $\overline{P}_{e,t}^{\text{max}}$ also reduces. However, if we keep the active power setpoint to a constant value, $\overline{P}_{e,t}^{\text{conv}}$ can be greater than the $\overline{P}_{e,t}^{\text{max}}$. In such a scenario, the converter will become unstable. Hence, it is important to determine appropriate power setpoints to ensure converter stability. The critical grid voltage for converter instability is determined by setting the active power setpoint equal to the maximum deliverable power in (24).

In the proposed approach, the EV aggregator discharge capacity is expressed in terms of the active and reactive power setpoints $P_{e,t}^{\text{EV}} = \sqrt{(P_{e,t}^{\text{set}})^2 + (Q_{e,t}^{\text{set}})^2}$. The active and reactive power setpoints replace the converter-to-grid active and reactive power in equation (24). Then, $\cos(\phi_{e,t}) = \frac{P_{e,t}^{\text{set}}}{P_{e,t}^{\text{EV}}}$ and rearranging (24), the critical grid voltage for instability is expressed as follows:

$$V_{e,t}^{\text{cr}} = \sqrt{\frac{2(P_{e,t}^{\text{set}})^2 x_g}{(P_{e,t}^{\text{set}})^2 + (Q_{e,t}^{\text{set}})^2 + Q_{e,t}^{\text{set}}}} \quad (25)$$

An example of the active and reactive power setpoints for varying grid voltages obtained using (25) is illustrated in Fig. 6. As the node voltage reduces, the active power setpoint decreases, and the reactive power setpoint increases to ensure converter stability. It can also be noticed that the node voltage at which the reactive power setpoint saturates to zero is larger for the case with larger $P_{e,t}^{\text{EV}}$, which indicates a stricter voltage security criterion for larger EV aggregators. Hence, to ensure the voltage security of EV aggregators, we treat the active and reactive power setpoints as decision variables and replace the aggregated discharge capacity $P_{e,t}^{\text{EV}}$ with the active power setpoint $P_{e,t}^{\text{set}}$. This change in the active power of EV aggregators is reflected in their frequency support capability.

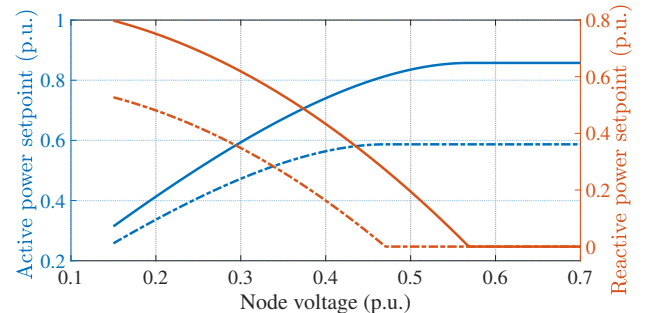


Fig. 6: Active and reactive power setpoints for $P_{e,t}^{\text{EV}} = 0.857$ p.u. (solid curves) and $P_{e,t}^{\text{EV}} = 0.586$ p.u. (dotted curves).

In this paper, in addition to the converter voltage security constraints, the EV active power setpoint is also constrained by the low-voltage ride-through (LVRT) requirements for grid voltage sags. Depending on the critical voltage for converter stability, there is a certain amount of time the converter must remain online and supply reactive power to the grid to ensure grid code compliance. The minimum ride-through time for the converter grid voltage taken from Table 12 of the IEEE Standard 2800-2022 [28] is illustrated in Fig. 7. Let the ride-through time be a function of the critical grid voltage in equation (25), then the minimum amount of EV aggregator energy required to satisfy the LVRT requirement is expressed as follows:

$$B_{e,t}^{\text{LVRT}} = \frac{P_{e,t}^{\text{set}} \times \mathbb{T}(V_{e,t}^{\text{cr}})}{\eta^{\text{EV},-}} \quad (26)$$

where $\mathbb{T}(V_{e,t}^{\text{cr}})$ is the minimum ride-through time corresponding to the critical grid voltage for converter instability (blue curve in Fig. 7). By setting the minimum energy required for LVRT to be greater than or equal to the lower capacity limit of EV aggregators, the LVRT constraint on $P_{e,t}^{\text{set}}$ is enforced.

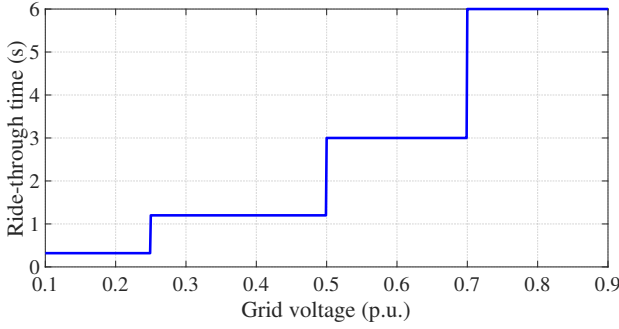


Fig. 7: Minimum ride-through times under grid voltage sag.

V. OPTIMIZATION FRAMEWORK FOR INTEGRATION LIMIT OF RENEWABLE ENERGY RESOURCES

In this paper, a NLP framework is developed to evaluate the RER penetration limit while keeping the frequency security constraints within limits. The framework also incorporates the classical ACOPF constraints such as the nodal power balance, generator, transmission line limits, and energy storage operation constraints. The NLP framework also incorporates the IR and PFR contribution from aggregated EVs while enforcing the converter voltage security and LVRT constraints. The optimization framework is structured as follows:

$$\min \left(\sum_{n \in \mathcal{N}} \lambda_c P_{n,t}^C + \sum_{g \in \mathcal{G}} c_g^f P_{g,t} + \sum_{s \in \mathcal{S}} (c_s^- P_{s,t}^- + c_s^+ P_{s,t}^+) \right) \quad (27)$$

subject to

$$P_{n,t} = \sum_{g \in \mathcal{G}_n} P_{g,t} + \sum_{p \in \mathcal{P}_n} P_{p,t} + \sum_{w \in \mathcal{W}_n} P_{w,t} + \sum_{s \in \mathcal{S}_n} (P_{s,t}^- + P_{s,t}^+) - \sum_{e \in \mathcal{E}_{a,n}} L_{e,t}^{\text{EV}} + P_{n,t}^C - P_{n,t}^D \quad (28a)$$

$$Q_{n,t} = \sum_{g \in \mathcal{G}_n} Q_{g,t} - Q_{n,t}^D \quad (28b)$$

$$P_{n,t} = V_{n,t} \sum_{i \in \mathcal{N}} V_{i,t} \left(G_{ni} \cos(\delta_{n,t} - \delta_{i,t}) + B_{ni} \sin(\delta_{n,t} - \delta_{i,t}) \right) \quad \forall n \in \mathcal{N} \quad (28c)$$

$$Q_{n,t} = V_{n,t} \sum_{i \in \mathcal{N}} V_{i,t} \left(G_{ni} \sin(\delta_{n,t} - \delta_{i,t}) - B_{ni} \cos(\delta_{n,t} - \delta_{i,t}) \right) \quad \forall n \in \mathcal{N} \quad (28d)$$

$$0 \leq P_{n,t}^C \leq P_{n,t}^D \quad \forall n \in \mathcal{N} \quad (28e)$$

$$V_n^{\min} \leq V_{n,t} \leq V_n^{\max} \quad \forall n \in \mathcal{N} \quad (28f)$$

$$u_{kl,t} P_{kl}^{\min} \leq V_{k,t}^2 |Y_{k,l}| \cos(\theta_{k,l}) - V_{k,t} V_{l,t} |Y_{k,l}| \times \cos(\delta_{k,t} - \delta_{l,t} - \theta_{k,l}) \leq u_{kl,t} P_{kl}^{\max} \quad \forall k, l \in \mathcal{L} \quad (28g)$$

$$u_{g,t} P_g^{\min} \leq P_{g,t} \leq u_{g,t} P_g^{\max} \quad \forall g \in \mathcal{G} \quad (28h)$$

$$u_{g,t} Q_g^{\min} \leq Q_{g,t} \leq u_{g,t} Q_g^{\max} \quad \forall g \in \mathcal{G} \quad (28i)$$

$$0 \leq P_{p,t} \leq P_{p,t}^{\max} \quad \forall p \in \mathcal{P} \quad (28j)$$

$$0 \leq P_{w,t} \leq P_{w,t}^{\max} \quad \forall w \in \mathcal{W} \quad (28k)$$

$$\overline{P_s^+} \leq P_{s,t}^+ \leq 0 \quad \forall s \in \mathcal{S} \quad (28l)$$

$$0 \leq P_{e,t}^- \leq \overline{P_s^-} \quad \forall s \in \mathcal{S} \quad (28m)$$

$$\underline{E}_s \leq E_{e,t-1} - \left(\frac{P_{s,t}^-}{\eta_s^-} + P_{s,t}^+ \eta_s^+ \right) \Delta t \leq \overline{E}_s \quad \forall s \in \mathcal{S} \quad (28n)$$

$$\sum_{p \in \mathcal{P}} P_{p,t} + \sum_{w \in \mathcal{W}} P_{w,t} + \sum_{s \in \mathcal{S}} P_{s,t}^d \leq (1 - \beta_t) \sum_{n \in \mathcal{N}} P_{n,t}^D \quad (28o)$$

$$\beta_t H_t^{\text{CV}} + \sum_{e \in \mathcal{E}_a} H_{e,t}^{\text{EV}} P_{e,t}^{\text{set}} \geq \frac{\Delta P_t}{2 \left| \frac{df}{d\tau} \right|_{\text{lim}}} \quad (28p)$$

$$\beta_t (R_t^T + D_t^{\text{CV}}) + \sum_{e \in \mathcal{E}_a} P_{e,t}^{\text{set}} R_{e,t}^{\text{EV},*} \geq \frac{\Delta P_t}{\Delta f_{\text{ss,lim}}} \quad (28q)$$

$$\frac{\Delta P_t}{\beta_t (R_t^T + D_t^{\text{CV}}) + \sum_{e \in \mathcal{E}_a} P_{e,t}^{\text{set}} R_{e,t}^{\text{EV},**}} \left(1 + e^{-\xi_t \omega_t \tau_t^{\max}} \times \sqrt{\frac{T_r \beta_t (R_t^T - F_t^T)}{2(\beta_t H_t^{\text{CV}} + \sum_{e \in \mathcal{E}_a} P_{e,t}^{\text{set}} H_{e,t}^{\text{EV}})}} \right) \leq \Delta f^{\text{lim}} \quad (28r)$$

$$H_{e,t}^{\text{EV}} \leq \frac{K_e^{\text{IR}} f_r}{2}, \quad \forall e \in \mathcal{E}_a \quad (28s)$$

$$H_{e,t}^{\text{EV}} \leq \frac{f_r \beta_t H_t^{\text{CV}}}{\Delta P_t}, \quad \forall e \in \mathcal{E}_a \quad (28t)$$

$$R_{e,t}^{\text{EV},*} \leq \frac{1}{R_e^{\text{EV}}}, \quad R_{e,t}^{\text{EV},**} \leq \frac{1}{R_e^{\text{EV}}}, \quad \forall e \in \mathcal{E}_a \quad (28u)$$

$$R_{e,t}^{\text{EV},*} \leq \frac{\beta_t (R_t^T + D_t^{\text{CV}}) + \sum_{i \in \mathcal{E}_a, i \neq e} P_{i,t}^{\text{set}} R_{i,t}^{\text{EV},*}}{\Delta P_t - P_{e,t}^{\text{set}}}, \quad \forall e \in \mathcal{E}_a \quad (28v)$$

$$R_{e,t}^{\text{EV},**} \leq \frac{\beta_t (R_t^T + D_t^{\text{CV}}) + \sum_{i \in \mathcal{E}_a, i \neq e} P_{i,t}^{\text{set}} R_{i,t}^{\text{EV},**}}{\rho \Delta P_t - P_{e,t}^{\text{set}}}, \quad \forall e \in \mathcal{E}_a \quad (28w)$$

$$\sqrt{\frac{2(P_{e,t}^{\text{set}})^2 x_g}{\sqrt{(P_{e,t}^{\text{set}})^2 + (Q_{e,t}^{\text{set}})^2} + Q_{e,t}^{\text{set}}}} \leq V_{e,t} - \Delta V_e, \quad \forall e \in \mathcal{E}_a \quad (28x)$$

$$\sqrt{(P_{e,t}^{\text{set}})^2 + (Q_{e,t}^{\text{set}})^2} \leq P_{e,t}^{\text{EV}}, \quad \forall e \in \mathcal{E}_a \quad (28y)$$

$$B_{e,t}^{\text{EV}} - \frac{P_{e,t}^{\text{set}} \times \mathbb{T}(V_{e,t} - \Delta V_e)}{\eta^{\text{EV},-}} \geq \underline{B}_{e,t}^{\text{EV}}, \quad \forall e \in \mathcal{E}_a. \quad (28z)$$

In the above formulation, objective (27) minimizes the load curtailment while simultaneously minimizing the generator fuel costs. The ESS charging and discharging costs c_s^-, c_s^+ are set such that ESS charge when the system generation is surplus and discharge when the total generation cannot cater to the system demand. Equations (28c) and (28d) enforce the equality constraints for nodal active and reactive power balance. Equations (28e) and (28f) denote the limits for load curtailment and node voltages. The transmission line limit constraint is enforced using equation (28g), while the generator limits are enforced using equations (28h)-(28k). The lower limits of conventional generators are relaxed to avoid integer variables in the NLP, which is a common procedure for solving ACOPF in planning problems [33]. It should also be noted that the generator and transmission line limits depend on their forced outage status $u_{g,t}$ and $u_{ij,t}$, which are determined and deployed during the stochastic simulations. The ESS operation constraints are enforced using equations (28l)-(28n).

In this paper, the RER limit is enforced using a linear constraint in equation (28o) that emphasizes that the RERs can only account for a certain percentage of system demand during time t . The value of β_t is majorly determined by the frequency security constraints in equations (28p), (28q), and (28r), which correspond to the maximum RoCoF, QSS frequency, and frequency nadir constraints, respectively. Depending on the outage statistics, a capacity flow model is solved to obtain the values of $\gamma_{g,t}$, which are subsequently used to determine $D_t^{\text{CV}}, H_t^{\text{CV}}, R_t^T, F_t^T$. Details of the capacity flow model is present in reference [34]. The capacity flow model is solved as a workaround for introducing integer variables within the NLP. Integer variables convert the NLP into a mixed-integer NLP (MINLP), significantly increasing computation times. While the optimization framework tries to minimize the value of β_t to minimize the fuel costs, the frequency security constraints impose a limit on the reduction of β_t , resulting in a dispatch that is frequency secure.

The e^{th} EV aggregator's contribution to frequency response is modeled using the setpoint active power and the EV inertia and droop reciprocals. The expression for the EV inertia constant given by equation (12) is expressed as constraints in (28s) and (28t). Similarly, the QSS and frequency nadir droop reciprocals in (17) and (19) are enforced using constraints (28u), (28v), and (28w). Using the expression for the critical voltage for converter stability from (25), constraint (28x) is formulated. Based on the grid voltage reduction ΔV_e , the power setpoints are controlled such that the critical operating voltage for the converter remains does not exceed $V_{e,t} - \Delta V_e$. The aggregator's power setpoints are dependent on the grid voltage $V_{e,t}$ at the node where the aggregator is connected. It can also be noticed that the grid voltage $V_{e,t}$ is a decision variable whose steady-state limits are enforced using constraint (28f). The active and reactive power setpoints for the converters are further constrained by the hourly discharge limit for the aggregator $P_{e,t}^{\text{EV}}$, obtained using equation (21). The LVRT constraints for EV aggregators are enforced as equation (28z). In this paper, for simplicity, the function $\mathbb{T}$ in constraint (28z)

is approximated by a quadratic polynomial fit of the minimum ride-through times in Fig. 7.

VI. SIMULATION AND RESULTS

The efficacy of the proposed approach is demonstrated using the RTS-GMLC test system [35]. This system is well suited for this work since it is one of the few available test systems that incorporate RERs, including PV, wind, concentrated solar power (CSP), hydro, and also ESSs. The RER resource database consists of day-ahead forecasts recorded at hourly intervals for a year. The generation mix consists of 74 conventional generation units, 20 hydro units, 25 PV farms, 31 roof-top PV, 1 CSP, 4 winds farms, and one ESS. There are 73 nodes distributed among three regions. Each area consists of distributed loads with a peak of 2850 MW.

The EV aggregator is assumed to be present in bus 313 of the RTS-GMLC system since the particular bus already includes an ESS. The traveling patterns of EV users are extracted using the national household travel survey (NHTS) database [32]. The total aggregated EV power for IR and PFR at each hour is obtained using Algorithm 1. This work considers high EV integration scenario with the total number of EV users being 100,000. Each EV has a battery with a rated charging/discharging capacity of 15 kW and an energy capacity of 60 kWh. Efficiencies for ESSs, individual EVs, and EV aggregators are taken as 0.85.

The generator inertia, fraction of power from high-pressure turbines, droop, and damping coefficient are extracted using the procedure in reference [36]. The maximum RoCoF values obtained in this paper are evaluated considering a 50 ms time delay of the RoCoF relay [30]. The NLP is solved using the IPOPT solver in pyomo [37]. The frequency security limits considered in this work are derived from Table 1 of the NERC standard BAL-003-1 for ERCOT, USA [38], and they are implemented as follows:

- Disturbance size (ΔP_t): The disturbance size is assumed to vary at all simulation hours, and it is considered to be 10% of the load demand during that hour. This disturbance size during peak demand roughly translates to the N-2 criterion for RTS-GMLC system, which is also the resource contingency criterion in BAL-003-1.
- Frequency nadir limits (Δf^{lim}): It is set as 0.7 Hz according to the prevailing under-frequency load shedding (UFLS) limit in BAL-003-1 for ERCOT.
- RoCoF limits ($\left| \frac{df}{dt} \right|^{\text{lim}}$): As reported by reference [39], under critical inertia, it takes 0.416 seconds for ERCOT's frequency to decrease from 59.7 Hz to 59.3 Hz, corresponding to the RoCoF of 0.96 Hz/s. To ensure that the system inertia remains above critical inertia, 0.96 is taken as the RoCoF limit in this work.
- QSS frequency limit ($\Delta f^{\text{ss,lim}}$): It is set as 50% of the frequency nadir limit.

A. Validation of the modified MM-SFR model

First, we have compared the performance of our proposed MM-SFR-based analytical model with IR and PFR constraints against the baseline simulation-based model and the virtual

synchronous machine (VSM)-based models in the literature. Three models are formulated as follows:

- Case I: The baseline model illustrated in Fig. 8 is simulated in MATLAB/Simulink and is used. The EV IR and PFR modules are directly incorporated with the base MMSFR model. This model considers filtering time constants and saturation blocks of IR and PFR modules. The IR gain $K_e^{\text{EV}} = 2$ and rated droop $R_e^{\text{EV}} = 0.01$ are used.
- Case II: The VSM strategy proposed in references [17]–[19] for emulating the frequency support of converter-based resources is integrated with the MMSFR model. Static values of virtual inertia constant (evaluated using (9)) and rated droop coefficient are used.
- Case III: Proposed MMSFR model with IR and PFR constraints is deployed. We use equations (12), (17), and (19) to dynamically update the inertia and droop reciprocals based on the varying disturbance sizes.

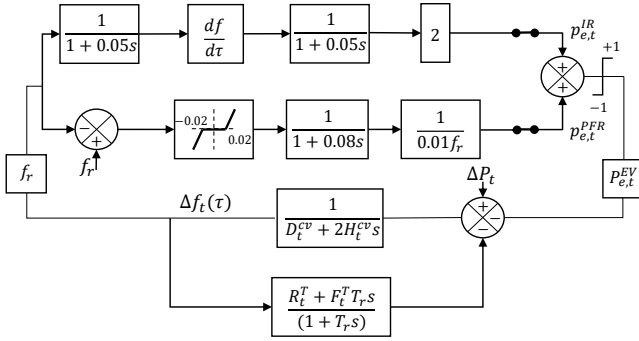


Fig. 8: Baseline model for validation of modified MM-SFR.

The performance of the three models for varying disturbance sizes and EV penetration is illustrated in Fig. 9. It is clear that the proposed MM-SFR model with IR and PFR constraints in case III has a better performance for all the frequency security metrics, especially as the disturbance sizes get more severe. In case II, the VSM models are reasonably accurate for small disturbance sizes, after which they incur high errors. It is due to the fact that the virtual inertia and droop values are at their rated values until the saturation blocks limit the IR and PFR power to 1 p.u. This effect is more severe when the EV penetration is higher, as seen in the dotted plots of Fig. 9. The proposed analytical model in case III successfully models the saturation blocks and accurately estimates the frequency security indices, even under the high penetration of EVs. These observations demonstrate that the proposed model can emulate the simulation-based baseline model in incorporating the frequency support from the EVs, while being analytically tractable to be included in the NLP framework.

B. One-day Dispatch Considering RER Penetration Limits

The variation in discharge capacity of EV aggregator for IR and PFR support is illustrated in Fig. 10. It can be observed that the discharge capability of the EV aggregator peaks during the evening. This occurs since the EVs mostly travel during the day, leading to low SoC by the evening when they are plugged into the grid for charging by the users. It can also be

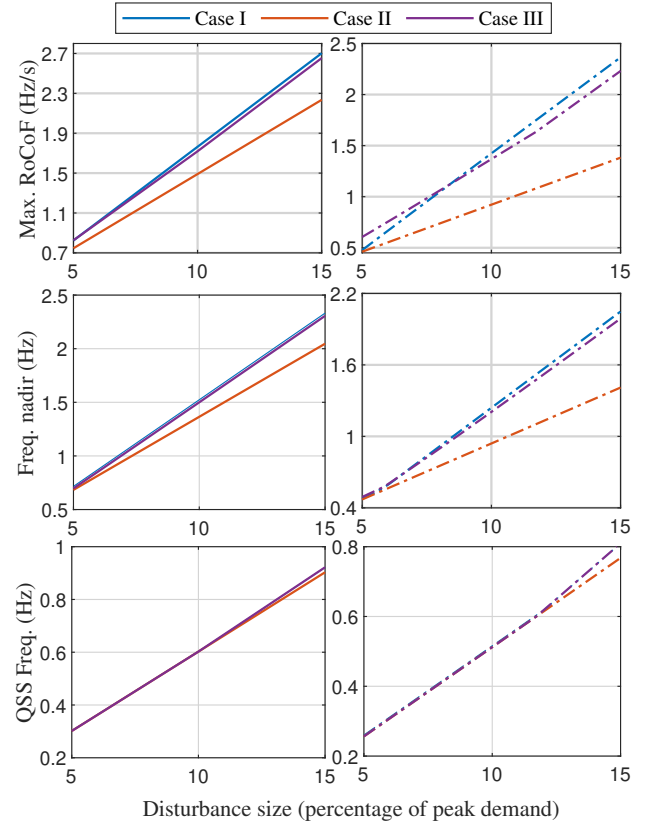


Fig. 9: Model validation results (solid lines represent 50 MW EV power while dotted lines represent 200 MW EV power.)

observed that EV discharge capacity increases along with the system generation, thus leading to the availability of a larger number of EVs for frequency support during the most stressful demand hours of the day. In addition, the increased load on the grid due to the charging of EVs leads to the dispatch of more generators for frequency support. This illustrates how the increased number of EVs in the system can be used to the advantage of operators to increase system frequency security.

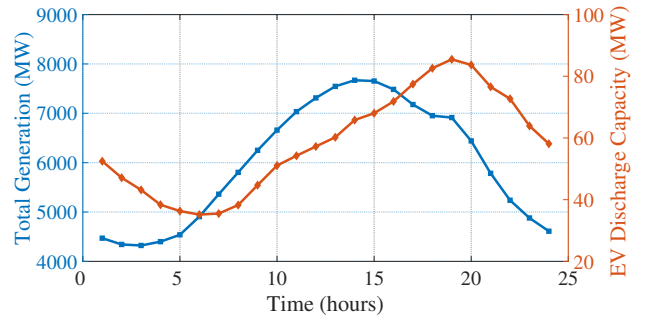


Fig. 10: 24-hour system dispatch and EV discharge capacity.

Next, a 24-hour case study is performed to illustrate the proposed method. Four cases are analyzed as follows:

- Case I: At every simulation hour, an ACOPF is simulated without considering the RER integration limit (constraints (28o)–(28z) are ignored).
- Case II: In this case, the NLP is solved while considering the frequency security constraints and RER integration factor.

However, the frequency support from EVs is not considered (constraints (28s)-(28z) are ignored and $P_{e,t}^{\text{set}} = 0$).

- Case III: In this case, the frequency support of EVs is modeled without considering the converter voltage security constraints i.e. constraints (28x)-(28z) are not considered and $P_{e,t}^{\text{set}} = P_{e,t}^{\text{EV}}$.
- Case IV: All the constraints in the NLP are considered. The voltage drop in EV aggregator terminals is 0.8 p.u.

In the abovementioned cases, simulations are performed for the peak demand day of the RTS-GMLC system, with frequency response and RER penetration assessed every hour. The frequency nadir, RER power, and β_t for the 24-hour dispatch are summarized in Fig. 11.

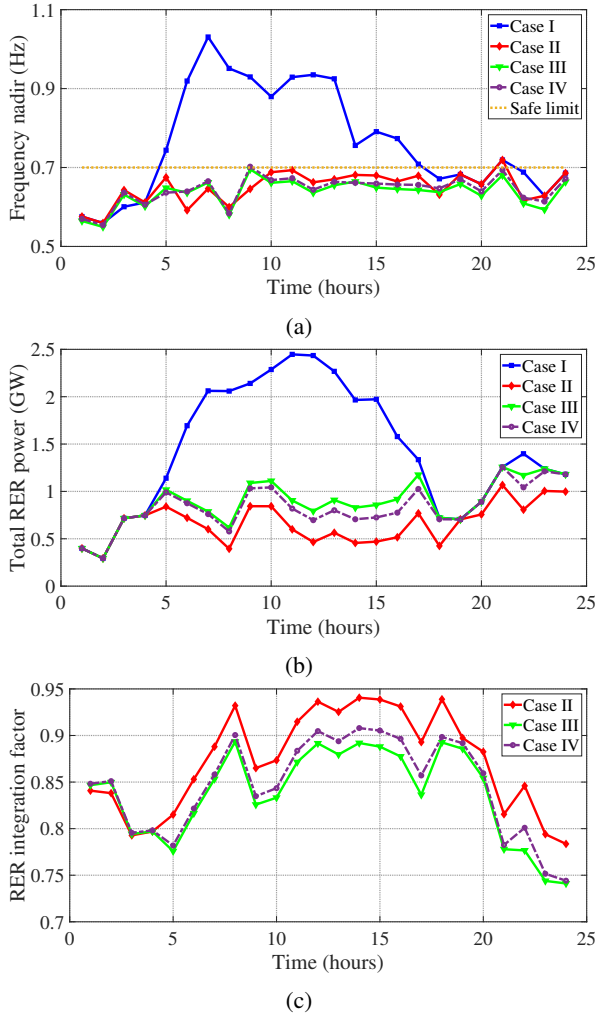


Fig. 11: a) System frequency nadir, b) Total RER generation, and c) β_t for the 24-hour dispatch.

The following can be observed from Fig. 11:

- In case I, Fig. 11a shows that the system experiences frequency nadir violations in multiple hours. This is because the system dispatch prioritizes RER integration due to the low production cost associated with these resources, as evident with the largest RER power for cases I in Fig. 11b. This causes the inertia, regulation, and other parameters from conventional generators to reach low values, affecting the frequency nadir. This effect is minimized in case II

where the RER integration limits are enforced. Using the RER integration factors in Fig. 11c, the RER penetration in the system is limited to satisfy the frequency nadir constraint. It should be mentioned that there is a minor nadir limit violation in hour 21 of case II. This is due to the linear constraint approximating the RER integration limit in (28o).

- The contribution of EVs in promoting the RER integration is illustrated by the case III plots in Fig. 11. When the frequency support from EVs is incorporated, the system experiences higher penetration of RERs between hours 5-16 while retaining a similar level of frequency security as case II. As seen in Fig. 11c, there is a significant reduction in β_t for case III during hours 5-16 compared to case II, contributing to larger RER power in Fig. 11b.
- Case IV illustrates the impacts of imposing the converter voltage security constraints on the RER penetration limit. As expected, when the voltage of the bus connected to the aggregated EVs drops to $V_{e,t} - \Delta V_e$, the active and reactive power setpoints are readjusted in proportion to the voltage drop. The reduced active power setpoint affects the frequency support from the EV aggregators and causes the system to impose heavier restrictions on the penetration of RERs to ensure frequency security. Due to this, the RER penetration is smaller in case IV compared to case III. This is also evident in Fig. 12 where the RER integration factor increases with increased ΔV_e . Neglecting the converter voltage security constraint causes an overestimation of the frequency support from energy storage devices in case III, leading to an optimistic estimation of the RER penetration limit. The proposed approach addresses this issue by including the voltage security constraint while determining the RER penetration limit.

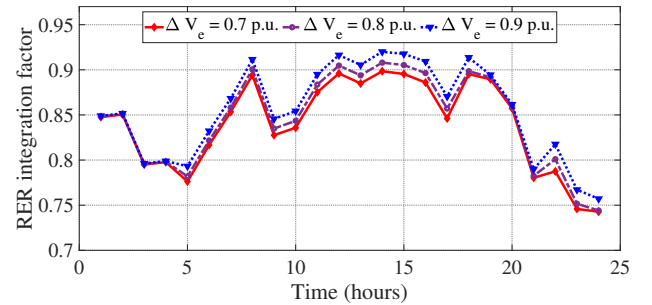


Fig. 12: RER integration limits for varying voltage sag levels.

Since the proposed NLP is non-linear and non-convex, the solution quality of the 24-hour dispatches generated by the NLP is analyzed using the overall unscaled NLP error in Fig. 13. The overall NLP error is the maximum among primal infeasibility (constraint violation), dual infeasibility, and complementarity, details of which are present in [40]. The NLP is solved using the overall error tolerance of $1e^{-4}$. However, the figure shows that the NLP errors are significantly smaller than the tolerance in all simulation hours, indicating that the problem constraints and optimality conditions are clearly satisfied. In addition, we also tested the possible global convergence of the algorithm with the “multistart” solver with “midpoint guess and bound” strategy. With 1000

iterations per dispatch hour in the “multistart” solver, the same dispatch results as Fig. 11 were obtained, indicating a strong convergence performance of the proposed NLP.

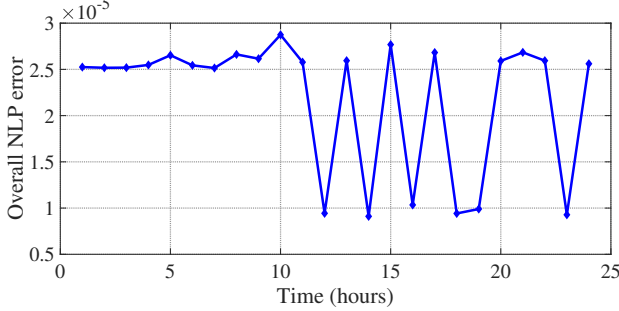


Fig. 13: Overall NLP error for 24-hour dispatch in case IV.

C. System Frequency Response and RER Penetration with Stochastic Simulations

In this section, stochastic simulations are performed to assess the RER penetration under different simulation scenarios. Stochastic simulations capture a vast range of uncertainties introduced by the random generator and transmission line outages, RER inputs, and EV user charging/discharging characteristics. We simulate 1000 sample paths for cases I-IV, with one sample path performing a yearly simulation. Within a sample path, the procedure in Algorithm 1 is initially run to obtain the EV aggregator load and discharge capability. Then, within the hourly simulation, the next event approach is used to simulate the forced outages of generators and transmission lines. Details of the next event approach are present in reference [34]. The RER integration and frequency security indices obtained from the NLP dispatch are recorded over all the simulation hours and sample paths.

Initially, the cumulative distributions of frequency security indices are presented in Fig. 14. Maximum RoCoF, frequency nadir, and QSS frequency limits are violated with a much higher probability for case I when the RER integration limit is not considered. In addition, the violations are extremely severe, up to three times the safety limit. In cases II-IV, the RER limits imposed prevent the frequency nadir and QSS frequency from reaching extreme values, while some values violate the safe limits due to the multiple forced outages of generators. The IR and PFR support from the EV aggregator reduces the probability of violating the safe limits in case III compared to case II, as they can provide frequency support during the forced outages of generators. However, the converter voltage security constraints undermine the frequency support from the EV aggregator, resulting in slightly increased frequency security violations in case IV compared to case III. This is illustrated by the case IV curves being slightly lower than the case III curves in Fig. 14.

Table I illustrates the impacts of imposing the RER integration limit constraint. In case II, the total energy over the simulation year from RERs reduces significantly compared to case I. It is because the only way for case II to keep the frequency security constraints within limits is by curtailing the RER power and increasing the number of conventional generators online. However, when frequency support from

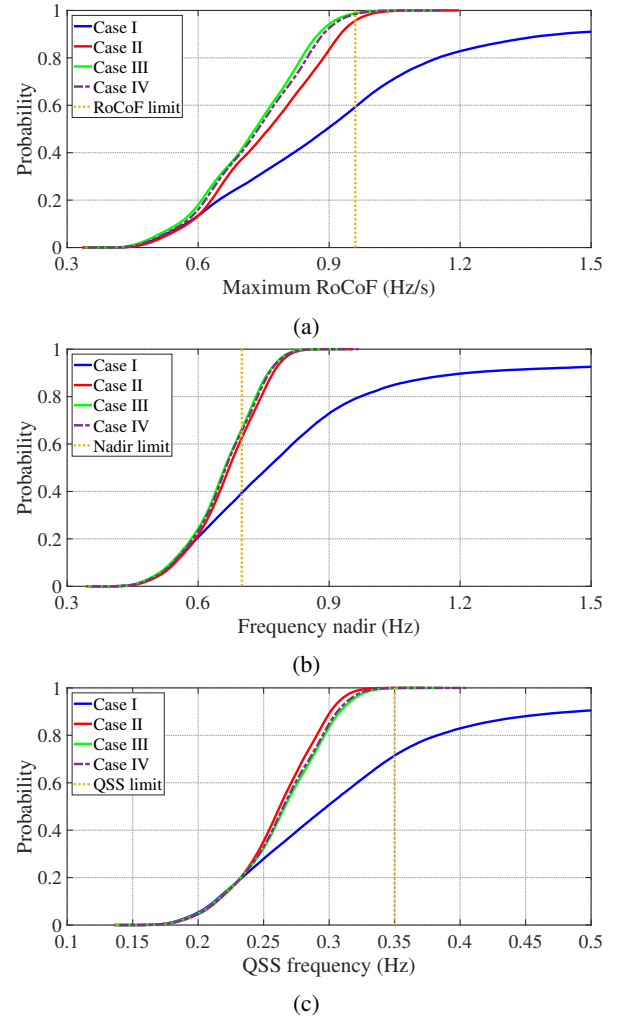


Fig. 14: Cumulative distributions of a) System maximum RoCoF, b) Frequency nadir, and c) QSS frequency.

EVs is considered, the RER energy improves considerably in cases III and IV compared to case II. This signifies that since the EV aggregators can assist in frequency support during a disturbance, fewer conventional generators must be online, and more RERs can be integrated without compromising frequency security. The computation times reported in Table I illustrate that the NLP in case II takes approximately twice the computation time of the default ACOPF, which is acceptable considering the NLP dispatches satisfy frequency security constraints. Also, it is observed that although incorporating the IR, PFR, and voltage security constraints increases the computational time in cases III and IV compared to case II, the increment is not significant.

TABLE I: RER integration results for all simulation cases.

Parameters	Case I	Case II	Case III	Case IV
RER energy (GWh/yr)	13385	6587.2	7953.8	7544.9
Computation time (s/step)	1.7431	3.0813	3.1638	3.1536

D. Discussion

Simulation results show that the proposed method can estimate the RER penetration limit in power grids while incorporating frequency support from EVs. Initially, the dynamic simulations illustrated the importance of the proposed dynamic virtual inertia and droop coefficients, especially under increased RER penetration. One-day dispatch simulations highlighted the efficacy of the proposed NLP in limiting the RER integration to satisfy the frequency security constraints. The simulations also highlighted how the frequency support from EVs can elevate the RER penetration in the system without compromising frequency security. Stochastic simulations highlighted the system frequency security and RER penetration under random forced outages of system generators.

The key benefit of the proposed NLP framework is that it is computationally inexpensive and suitable for use in planning frameworks. This is primarily attributed to the approach not requiring integer decision variables, removing the need to solve a MINLP problem. Regarding the applicability of the proposed framework in real-world, as mentioned in the ACOPF white paper issued by FERC in reference [41], “The ACOPF problem must be solved weekly in 8 hours, daily in 2 hours, hourly in 15 minutes, every five minutes in 1 minute and for self-healing post-contingency in 30 seconds.” The computation times reported in Table I illustrate that the proposed NLP adheres to these limits for very short periods (such as 5-minute dispatch) and planning problems (hourly resolution). In this work, we have primarily focused on solving the NLP hourly within a stochastic simulation to determine the RER penetration in the system. However, the computation times in Table I indicate the applicability of the NLP for shorter time scales.

VII. CONCLUSION

This paper formulates an improved approach to incorporate frequency support of EVs to promote RER integration. The contribution of EV aggregators to system frequency response is modeled using the dynamic virtual inertia and droop coefficients, which in turn are determined using the IR and PFR control modules. The resulting EV aggregator parameters are then deployed in the modified multi-machine system frequency response (MM-SFR) model to obtain the frequency security indices. To determine the discharge limits of EV aggregators for frequency support, a stochastic framework is developed. A non-linear optimization framework is then devised to determine the maximum penetration limit of RERs in the system. The framework incorporates frequency security constraints, IR and PFR constraints of EV aggregators, converter voltage security, and LVRT constraints. Simulation results on the IEEE RTS-GMLC illustrate that the proposed framework incorporates the IR and PFR contribution of EVs and improves the system frequency response and RER integration while also illustrating the importance of considering voltage security constraints. The proposed model can be used in system planning studies such as reliability analysis since the method does not require simulation models. Computation times show that the model can also help system operators determine the

RER penetration limits in near real-time with changes in EV penetration and system demand.

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